

EpigeneticsBench: Verifiable Evaluation of AI Agents on Epigenomics Analysis

Short-horizon tasks from real epigenomics workflows

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ABSTRACT

We introduce EpigeneticsBench, a verifiable benchmark for short-horizon epigenomics analysis. Rather than asking agents to produce open-ended biological interpretations, EpigeneticsBench evaluates whether they can execute focused analysis decisions from realistic workflow states and return deterministically gradable answers. The updated benchmark includes 106 evaluations across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation workflows. Across 5,088 parsed trajectories from 16 model-harness pairs, no system solves a majority of attempts: GPT-5.5 / Pi leads at 151/318 attempts (47.5%), followed by GPT-5.5 / OpenAI Codex at 135/318 attempts (42.5%) and GPT-5.4 / Pi at 134/318 attempts (42.1%). Difficulty is structured by assay source and workflow convention more than by nominal horizon label, and component-level scoring reveals frequent partial recovery despite endpoint failure. These results suggest that current agents can often locate and manipulate the right biological evidence, but still fail at brittle local decisions involving metric definitions, comparator scope, normalization state, and answer-surface constraints.

Topline Benchmark Performance

We ran the benchmark across 16 model-harness pairs. Pi denotes the Pi terminal coding harness. Passing requires exact recovery of the graded structured answer for an evaluation attempt. The updated manifest contains 5,088 valid result JSONs in a balanced three-replicate design: 106 evaluations for each of 16 model-harness pairs.

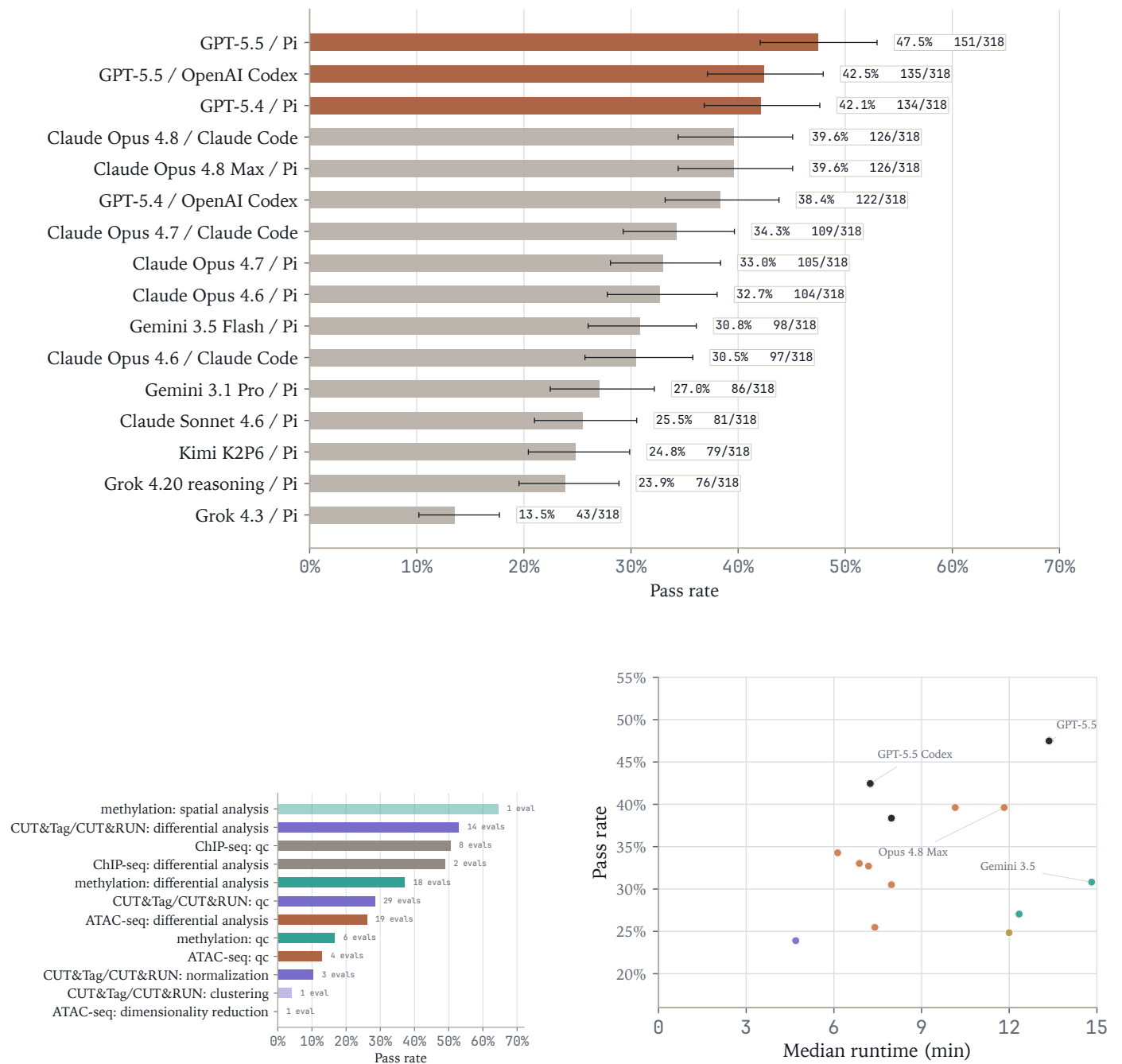


Figure 1: Topline benchmark performance. Top: run-level pass rate by model-harness pair across all parsed attempts. Bottom left: pass rate by assay and task family. Bottom right: pass rate against median runtime. Wilson confidence intervals and evaluation-level replicate statistics are reported in Table 2.

Introduction

Epigenomics connects regulatory sequence, chromatin accessibility, histone modification, DNA methylation, and genome organization to cell state and gene regulation. Modern assays can measure these layers at population, single-cell, and spatial resolution, but the resulting analyses are brittle: small choices in genome build, peak filtering, motif background selection, sample aggregation, or denominator definition can change a biological conclusion [1, 2, 3, 4].

AI agents can now inspect files, write analysis code, run command-line tools, and return structured biological summaries. Benchmarks have therefore moved from question answering toward executable scientific tasks. SpatialBench tests localized spatial biology analysis, scBench tests single-cell workflows, and SpatialBench-Long tests recovery of complex scientific conclusions from raw or near-raw spatial data

Benchmark Design

The benchmark contains 106 evaluations derived from practical epigenomics analysis workflows. Tasks are short-horizon: they isolate one focused analysis decision while preserving enough real data context that a model must inspect files, choose the relevant representation, and compute or verify the requested result. The benchmark follows the SpatialBench convention of withholding the full task set to reduce contamination while reporting aggregate model results and releasing representative examples.

Benchmark Construction Workflow

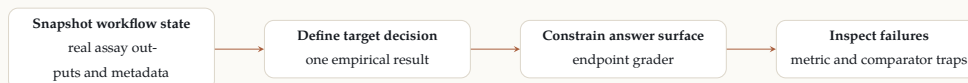


Figure 2: Benchmark construction workflow. Short-horizon epigenomics evaluations are built by selecting real analysis snapshots, defining a single verifiable target decision, constraining the final answer schema, and using trajectory inspection to identify shortcut-prone or under-specified tasks.

The benchmark is organized around analysis states rather than independent datasets. A single source study or workflow can contribute multiple evaluations that vary the target statistic, comparator, or biological claim. This mirrors how working epigenomics analyses proceed: a scientist repeatedly queries the same assay context to answer adjacent but distinct questions.

Tasks are designed to specify *what* should be recovered rather than *how* it should be recovered. Some procedural evaluations name an operation directly, but most scientific evaluations leave room for the agent to choose reasonable commands, scripts, and summary logic. This prevents the prompt from becoming a hidden solution path while keeping the answer surface verifiable.

[5, 6, 7].

Epigenomics needs a complementary short-horizon benchmark. Many consequential failures occur in compact analysis decisions rather than in full-project autonomy: deciding which count layer to use, whether a statistic should be computed per peak, per gene, or per sample, which contrast is biologically appropriate, or how a CpG, motif, or chromatin-state summary should be normalized.

This benchmark evaluates those local empirical decisions. Each task snapshots a realistic analysis state immediately before a target result. The agent must interact with the provided data and return a structured answer that can be graded deterministically. The benchmark is therefore not a test of broad regulatory-biology plausibility. It asks whether an agent can produce the specific verifiable result supported by the data, assay context, and requested answer surface.

The updated inventory contains 97 scientific evaluations and 9 procedural evaluations. Most graders are numeric tolerance checks (97/106 evaluations), with 9 all-of graders for compound structured answers. Component field scores are retained for diagnostics, but run-level pass/fail is the benchmark score. This distinction matters because some grader score fields are not normalized across all evaluations, whereas pass/fail has a consistent interpretation.

Manual trajectory inspection is used in the same spirit as SpatialBench and SpatialBench-Long: not to replace deterministic endpoint grading, but to make failure modes legible. The current metadata marks recurring traps such as wrong comparator or baseline, feature-namespace mismatch, wrong data layer, metric-definition mismatch, and prior-driven biological overinterpretation.

Motivating Example: One Local Decision Can Change the Answer

Short-Horizon Epigenomics Chokepoints

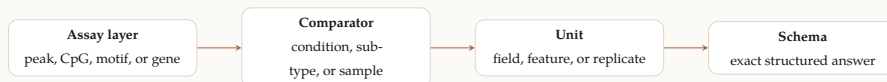


Figure 3: Short-horizon epigenomics chokepoints. A representative evaluation snapshots an analysis immediately before a local decision. Agents must identify the relevant assay layer, choose the requested comparison, compute the target statistic over the correct unit of analysis, and return a constrained answer that can be graded deterministically.

The short-horizon setting is easiest to see through a single evaluation. One representative ATAC-seq task begins after peak-level outputs and sample metadata have already been staged. The agent is not asked to reconstruct an entire study; it must choose the correct representation, comparator, denominator, and final answer schema for one constrained empirical result.

This kind of task is small enough to be verifiable but still scientifically fragile. A trajectory can inspect the right files and compute a plausible statistic, yet fail by using the wrong fea-

ture namespace, aggregating over the wrong biological unit, or substituting a nearby biological contrast for the requested one. Conversely, a passing answer requires more than memorized regulatory biology: it requires matching the requested output to the data layer and comparison actually supplied in the task.

This example illustrates why deterministic endpoint grading and diagnostic metadata are both useful. The benchmark score remains binary, because the target answer is a constrained scientific artifact. The failure analysis asks which local decision broke when the endpoint answer was wrong.

Evaluation Inventory

The benchmark covers four assay labels and six task families. CUT&Tag/CUT&RUN tasks are drawn from zebrafish chromatin workflow snapshots and emphasize quality control, differential chromatin or expression summaries, normalization, and one clustering task. ATAC-seq tasks cover B-ALL and cold-response analyses, while ChIP-seq tasks focus on quality-control and differential histone-mark summaries. Methylation tasks are dominated by GSE149608 and GSE149609 outputs, including CpG reports, coverage tables, noncanonical methylation checks, and differential methylation summaries.

Assay label	Evals	Sources	Task families	Horizon
CUT&Tag/CUT&RUN	47	Zebrafish chromatin workflow snapshots	QC (29), differential analysis (14), normalization (3), clustering (1)	Small (47)
ATAC-seq	24	B-ALL ATAC, cold-response ATAC	Differential analysis (19), QC (4), dimensionality reduction (1)	Small (24)
ChIP-seq	10	ChIP-seq workflow snapshots	QC (8), differential analysis (2)	Small (9), long (1)
DNA methylation	25	GSE149608, GSE149609	Differential analysis (18), QC (6), spatial analysis (1)	Long (25)

Table 1: Evaluation inventory. The short-horizon epigenomics benchmark groups evaluations by assay label while varying the source workflow, task family, and expected answer surface within each group.

Across all 106 evaluations, 47 are QC tasks, 53 are differential-analysis tasks, 3 are normalization tasks, and one each tests dimensionality reduction, clustering, and spatial analysis. The updated result manifest contains 5,088 valid trajectories from 16 model-harness pairs, with exactly 318 attempts per model-harness pair.

Results

Endpoint grading identifies a leading tier rather than a fully ordered leaderboard

Across 16 model-harness pairs and 5,088 valid trajectories, deterministic final-answer grading produced 1,672 passes (32.9%). Success was concentrated but not cleanly separated across the leading systems. GPT-5.5 / Pi achieved the highest pass rate, passing 151/318 attempts (47.5%; Wilson 95% confidence interval (CI), 42.1–53.0). GPT-5.5 / OpenAI Codex

followed at 135/318 attempts (42.5%; 37.1–47.9), while GPT-5.4 / Pi reached 134/318 attempts (42.1%; 36.8–47.6). Claude Opus 4.8 Max / Pi and Claude Opus 4.8 / Claude Code both passed 126/318 attempts (39.6%).

The updated manifest is balanced: each model-harness pair contributes three rounds for each of 106 evaluations. The middle panel of Figure 4 therefore matches the full parsed set rather than a smaller sensitivity subset. The practical interpretation is unchanged from the earlier draft but sharper: GPT-5.5 / Pi is the descriptive leader, while the nearby systems form a leading tier whose Wilson intervals overlap.

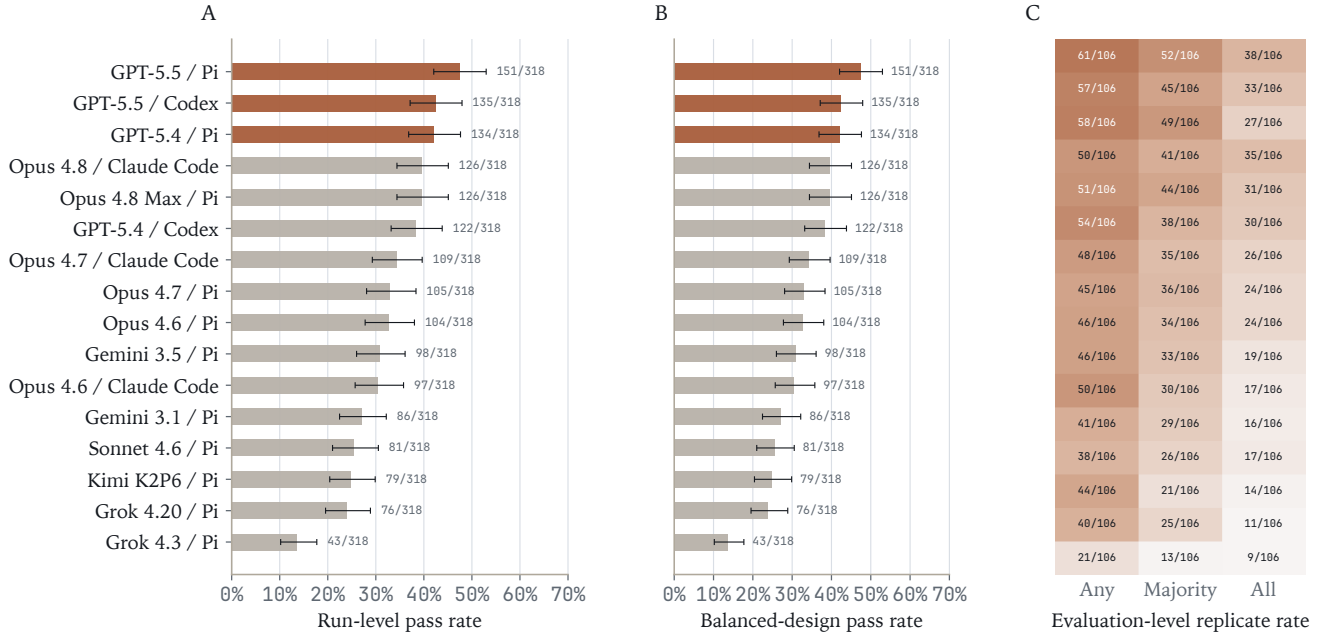


Figure 4: Endpoint success across balanced model-harness coverage. Left: pass rate by model-harness pair across all 5,088 parsed attempts. Middle: the same balanced 318-attempt design displayed as a shared-instance check. Right: evaluation-level replicate statistics, reporting the fraction of attempted evaluations with at least one passing attempt, a majority of passing attempts, or all available attempts passing. Wilson intervals summarize run-level uncertainty and do not account for evaluation-level clustering.

Model / harness	Pass runs	Pass %	Wilson 95% CI	Any	Majority	All	Cost	Median min
GPT-5.5 / Pi	151/318	47.5	42.1–53.0	61/106	52/106	38/106	\$1.284	13.4
GPT-5.5 / OpenAI Codex	135/318	42.5	37.1–47.9	57/106	45/106	33/106	–	7.2
GPT-5.4 / Pi	134/318	42.1	36.8–47.6	58/106	49/106	27/106	\$0.924	15.3
Claude Opus 4.8 Max / Pi	126/318	39.6	34.4–45.1	50/106	41/106	35/106	–	11.8
Claude Opus 4.8 / Claude Code	126/318	39.6	34.4–45.1	51/106	44/106	31/106	–	10.2
GPT-5.4 / OpenAI Codex	122/318	38.4	33.2–43.8	54/106	38/106	30/106	–	8.0
Claude Opus 4.7 / Claude Code	109/318	34.3	29.3–39.7	48/106	35/106	26/106	–	6.1
Claude Opus 4.7 / Pi	105/318	33.0	28.1–38.4	45/106	36/106	24/106	–	6.9
Claude Opus 4.6 / Pi	104/318	32.7	27.8–38.0	46/106	34/106	24/106	\$0.620	7.2
Gemini 3.5 Flash / Pi	98/318	30.8	26.0–36.1	46/106	33/106	19/106	\$1.587	14.8
Claude Opus 4.6 / Claude Code	97/318	30.5	25.7–35.8	50/106	30/106	17/106	–	8.0
Gemini 3.1 Pro / Pi	86/318	27.0	22.5–32.2	41/106	29/106	16/106	\$0.768	12.3
Claude Sonnet 4.6 / Pi	81/318	25.5	21.0–30.5	38/106	26/106	17/106	\$0.609	7.4
Kimi K2P6 / Pi	79/318	24.8	20.4–29.9	44/106	21/106	14/106	\$0.460	12.0
Grok 4.20 reasoning / Pi	76/318	23.9	19.5–28.9	40/106	25/106	11/106	\$0.280	4.7
Grok 4.3 / Pi	43/318	13.5	10.2–17.7	21/106	13/106	9/106	\$0.033	0.7

Table 2: Deterministic final-answer performance. Run-level pass rates are computed over the balanced updated manifest. Any, Majority, and All report evaluation-level replicate statistics over 106 evaluations for each model-harness pair. Cost is shown only when it is present for all attempts from that model-harness pair; dashes indicate incomplete or unavailable cost metadata. Median minutes summarizes the recorded result duration.

Assay source stratifies difficulty more clearly than horizon labels

Performance varied more by assay source and workflow provenance than by the coarse horizon label. ChIP-seq tasks passed at 241/480 attempts (50.2%), followed by CUT&Tag/CUT&RUN at 768/2,256 (34.0%), methylation at 400/1,200 (33.3%), and ATAC-seq at 263/1,152 (22.8%). Source-level summaries exposed a sharper split within these assay labels: GSE149608 methylation tasks passed at 388/1,152 attempts (33.7%), whereas the single GSE149609 task passed at 12/48 (25.0%). Cold-response ATAC tasks passed at 147/576 (25.5%), while B-ALL ATAC-prefix tasks passed at 116/576 (20.1%).

Task family alone was less informative. Differential-analysis tasks passed at 962/2,544 attempts (37.8%), while QC tasks passed at 662/2,256 (29.3%). However, the same nomi-

nal task family behaved differently across assay contexts: CUT&Tag/CUT&RUN differential-analysis tasks passed at 356/672 attempts (53.0%), while ATAC-seq differential-analysis tasks passed at 238/912 (26.1%). Agents are therefore not simply failing broad task types; they are sensitive to the representation, comparator, and denominator conventions of particular epigenomics workflows.

The horizon label should be interpreted cautiously. Long-horizon evaluations passed at 429/1,248 attempts (34.4%) versus 1,243/3,840 (32.4%) for small-horizon evaluations, but the long group is almost entirely methylation: all 25 methylation evaluations are long horizon, whereas all CUT&Tag/CUT&RUN and ATAC-seq evaluations are small horizon and only one ChIP-seq evaluation is long horizon (Figure 5). The small long-horizon advantage is consequently a source-composition effect rather than evidence that longer tasks are intrinsically easier.

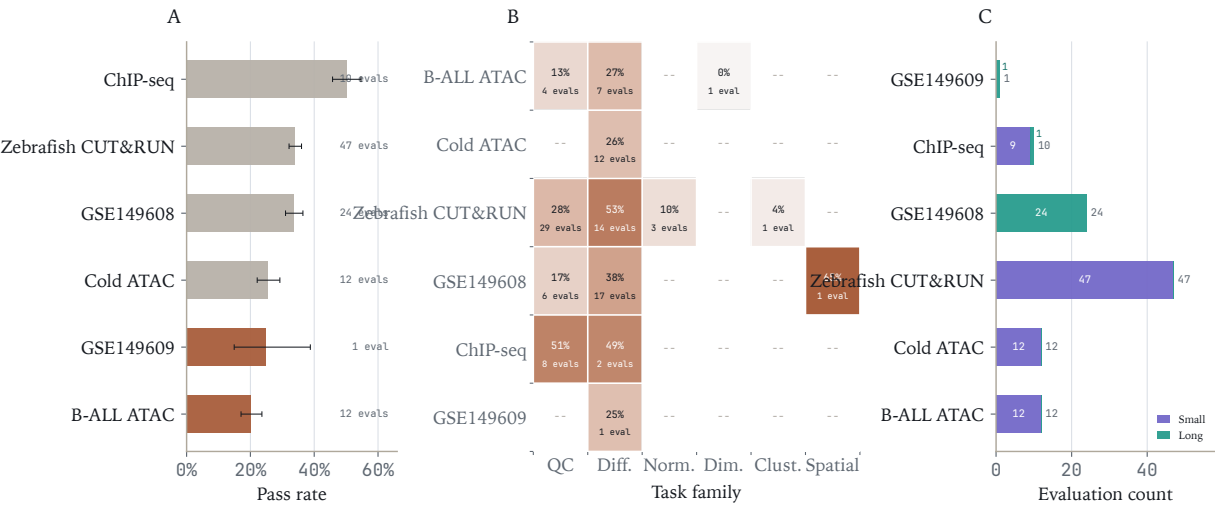


Figure 5: Performance varies by source workflow and is confounded with horizon. Left: pass rate by source prefix with run-level Wilson intervals. Middle: source-by-task pass-rate heatmap; cell text reports pass rate and evaluation count. Right: horizon composition by source prefix. Long-horizon evaluations are mostly methylation/GSE149608, so the higher long-horizon pass rate should not be interpreted as evidence that longer tasks are intrinsically easier.

Replicate statistics distinguish occasional recovery from stable recovery

Run-level pass rates are useful for ranking systems, but evaluation-level replicate statistics better capture reliability. GPT-5.5 / Pi passed at least one attempt on 61/106 evaluations, passed a majority of attempts on 52/106, and passed all three attempts on 38/106. GPT-5.5 / OpenAI Codex showed a similar drop from 57/106 any-pass evaluations to 45/106 majority-pass and 33/106 all-pass evaluations. In contrast, Grok 4.3 / Pi passed at least one attempt on 21/106 evaluations and a majority on 13/106.

This gap matters because short-horizon tasks are not purely deterministic from the agent’s perspective. A model can know the relevant assay logic but choose the wrong field name, denominator, or comparator in one replicate. Conversely, a single passing replicate does not imply stable operational competence. We therefore treat any-pass rates as evidence of reachable capability and majority/all-pass rates as stronger evidence of repeatable recovery.

Hard evaluations often reflect component conjunction failures

Evaluation difficulty was highly skewed. Across 106 evaluations, 21 had no passing attempts and 18 more had pass rates below 10%, whereas no evaluation was passed by every available attempt and only three exceeded 90% pass rate. The hardest evaluations were not uniformly cases of no analysis progress: 17 of 21 zero-pass evaluations had nonzero mean grader score, and 9 of 21 had field-level pass rates of at least 50%.

Field scores show that many failures occur at the final conjunction of components. Across all 5,088 runs, endpoint pass/fail was exactly equivalent to all component fields passing. The weighted field-level pass rate was 69.6%, while endpoint pass rate was 32.9%. This gap was largest for ATAC-seq dimensionality reduction, CUT&Tag/CUT&RUN normalization, ATAC-seq QC, CUT&Tag/CUT&RUN clustering, CUT&Tag/CUT&RUN QC, and ATAC-seq differential-analysis tasks. These results indicate that many agents recover substantial local structure but miss one required field, comparator, denominator, or answer-surface constraint.

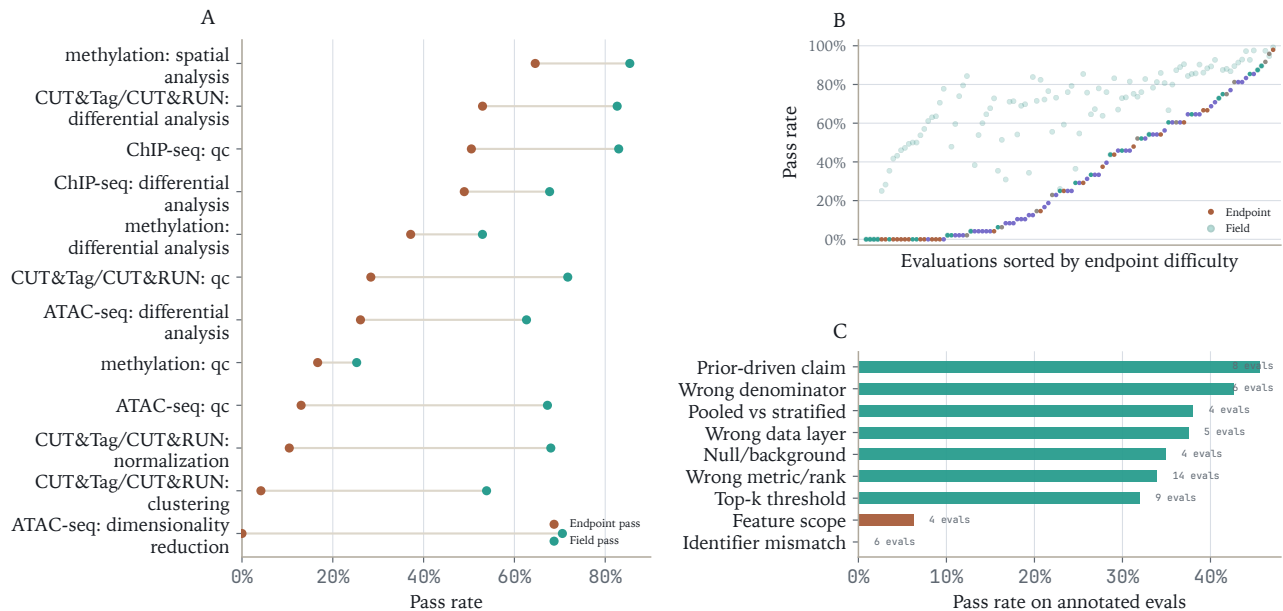


Figure 6: Endpoint failures often contain substantial partial progress. Left: endpoint pass rate and field-level pass rate by assay and task family. Top right: evaluation-level difficulty spectrum, with endpoint pass rate and field-level pass rate shown for each evaluation. Bottom right: annotated failure-mode severity; dot size indicates the number of uniquely annotated evaluations. Field scores and failure annotations are diagnostics, not replacement benchmark scores.

Failure annotations localize errors to concrete analysis decisions

Failure annotations support the component-conjunction interpretation. The most frequent annotated families were metric/statistic mismatch, normalization or representation mismatch, biological-interpretation errors, subset or comparator errors, and threshold-resolution instability. These categories match common epigenomics failure modes: reaching the right data object but summarizing the wrong layer, computing a plausible statistic with the wrong denominator, or substituting a nearby biological contrast for the requested comparison.

Annotation coverage is incomplete. Only 25/106 evaluations currently have failure-mode labels, including 10/21 zero-pass evaluations. Failure modes are also evaluation-level annotations rather than per-run observed labels, and co-annotated evaluations appear in multiple rows of the failure-mode summary. We therefore treat the taxonomy as a diagnostic sample of recurring traps rather than a complete decomposition of benchmark error.

Runtime failures are not the dominant error mode

Among parsed result JSONs, no trajectory reports a timeout or detected out-of-memory event. Recorded runtimes nevertheless vary widely: the median duration is 506 seconds and the 90th percentile is 2,351 seconds. Grok 4.3 / Pi is very fast, with a median duration of 41 seconds and a 13.5% pass rate. GPT-5.5 / Pi is slower, with a median duration of 802 seconds and the highest pass rate. These measurements should be interpreted as operational diagnostics rather than biological capability scores. Cost metadata are available for 2,875/5,088 attempts and are complete for only a subset of model-harness pairs, so they should not be used for a global cost-efficiency ranking in the current manuscript.

Discussion

The short-horizon epigenomics benchmark occupies a middle ground between localized workflow execution and long-horizon scientific reasoning. It is narrower than SpatialBench-Long because each task targets a focused result, but it is more empirical than biological question answering because the answer depends on interacting with real assay outputs.

The main result is not that agents fail universally. Leading systems recover roughly half of available endpoint answers, and many failures show substantial component-level progress. The limitation is reliability: even strong model-harness pairs remain sensitive to representation, metric definition, and comparator choice. These are precisely the local decisions that determine whether an epigenomics analysis is trustworthy.

The benchmark has several limitations. The updated

manifest is balanced across model-harness pairs, but the task inventory is still not balanced across assay families: CUT&Tag/CUT&RUN and methylation contribute more evaluations than ChIP-seq, and QC plus differential-analysis tasks dominate the inventory. Finally, deterministic graders intentionally constrain the answer surface; they are useful for measurement, but they do not capture every scientifically valid analysis path.

Methods

Evaluation construction

Candidate evaluations were derived from real epigenomics analysis workflows. An evaluation was retained when the target result could be reproduced from the provided data, expressed through a constrained final answer, and graded deterministically. Tasks were revised or removed when prompts over-specified the method, when reasonable analysis choices produced incompatible answers, or when the grader could not distinguish a supported result from plausible but unsupported biological output.

Agent runs and aggregation

Each model-harness pair was run with up to three replicates per evaluation. The updated manifest used here is `epibench_manifest_updated.csv`; the full SHA-256 hash is recorded in the derived summary facts table. All 5,088 listed result JSONs were downloaded and parsed successfully. The manifest contains a complete three-replicate design for 106 evaluations and 16 model-harness pairs. Failed, malformed, or nonpassing endpoint grader outputs are counted as nonpassing attempts when present; no downloaded result JSON was missing or invalid in this analysis.

Statistical analysis

The primary metric is run-level pass rate. Wilson binomial confidence intervals summarize run-level uncertainty and do not account for evaluation-level clustering. We therefore also report evaluation-level replicate summaries: whether a model-harness pair passed any attempt, a majority of attempts, or all three attempts for an evaluation. Because the updated manifest is balanced, run-level and balanced-design summaries are identical. Component field scores and failure-mode labels are reported as diagnostics rather than benchmark scores.

Data availability

Results files, public example evaluations, and representative trajectories will be made available at <https://github.com/latchbio/epigenetics-short-horizon-benchmark>. The derived summary tables used for this manuscript are in

References

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